

Setting up

First you'll import the parts of Python that you need for this project. Then you'll set things up so that you're ready to write the main functions for the game.

1

Create a file

Open IDLE and create a new file. Save it as "egg_catcher.py".



2

Import the modules

Egg Catcher uses three modules: **itertools** to cycle through some colours; **random** to make the eggs appear in random places; and **Tkinter** to animate the game by creating shapes on the screen. Type these lines at the top of your file.

```
from itertools import cycle
from random import randrange
from tkinter import Canvas, Tk, messagebox, font
```

The code only imports the parts of the modules that you need.

3

Set up the canvas

Add this code beneath the import statements. It creates variables for the height and width of the canvas, then uses them to create the canvas itself. To add a bit of scenery to your game, it draws a rectangle to represent some grass and an oval to represent the sun.

```
from tkinter import Canvas, Tk, messagebox, font
```

```
canvas_width = 800
```

```
canvas_height = 400
```

```
root = Tk()
```

```
c = Canvas(root, width=canvas_width, height=canvas_height, \
background='deep sky blue')
```

```
c.create_rectangle(-5, canvas_height - 100, canvas_width + 5, \
canvas_height + 5, fill='sea green', width=0)
```

```
c.create_oval(-80, -80, 120, 120, fill='orange', width=0)
```

```
c.pack()
```

The canvas will be sky blue and measure 800 x 400 pixels.

Use a backslash character if you need to split a long line of code over two lines.

This creates a window.

This creates the grass.

The **pack()** function tells the program to draw the main window and all its contents.

This line creates the sun.

4

See your canvas

Run the code to see how the canvas looks. You should see a scene with green grass, a blue sky, and a bright sun. If you feel confident, try to make your own scenery with shapes of different colours or sizes. You can always go back to the code above if you run into problems.

